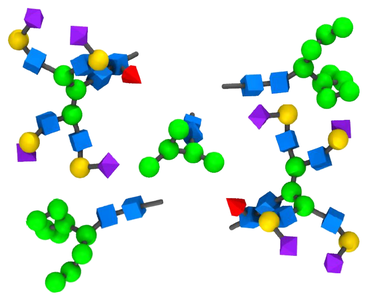
GlyGen

Computational and Informatics Resources for Glycoscience

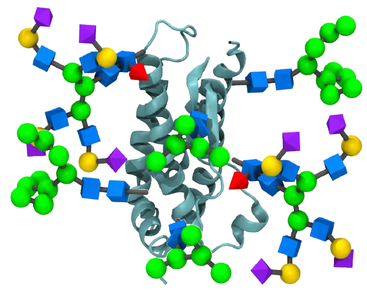
The GlyGen Consortium

UNIVERSITY OF GEORGIA · THE GEORGE WASHINGTON UNIVERSITY WASHINGTON, DC · GEORGETOWN UNIVERSITY · EMBL-EBI · UNIVERSITY OF DELAWARE

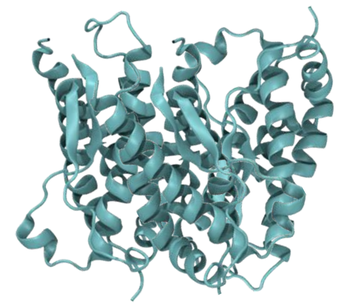
GlyGen Data



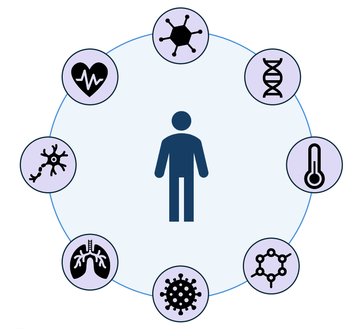
Glycan
Glycan data in GlyGen includes the lists of organisms in which the glycan was reported, motifs present in the glycan, relevant bio-synthetic enzymes, proteins that carry the glycan, proteins that bind the glycan, and supporting papers.



Site
Protein sites in GlyGen can have information about attached glycan structures, glycosylation, mutation (single nucleotide variation), phosphorylation, glycation, and mutagenesis information.



Protein
Protein data in GlyGen includes the lists of modified sites, attached and bound glycans, reported function and GO annotations, iso-forms and homologs, expression in different tissues and diseases, and supporting papers.

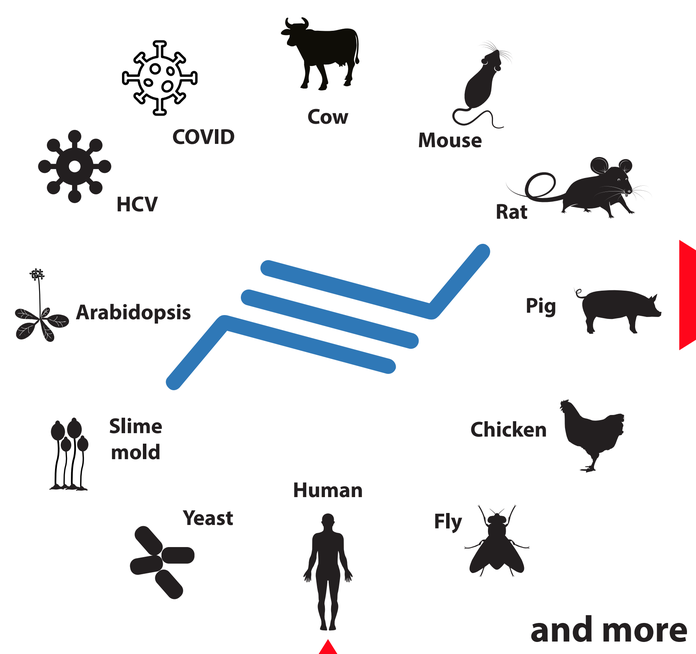


Biomarker
GlyGen contains data on glycan, glycoprotein and protein biomarkers that are used to predict, diagnose or monitor diseases. Biomarker data includes the biomolecules, disease, biomarker type, supporting evidence and associated publications.

Data integration

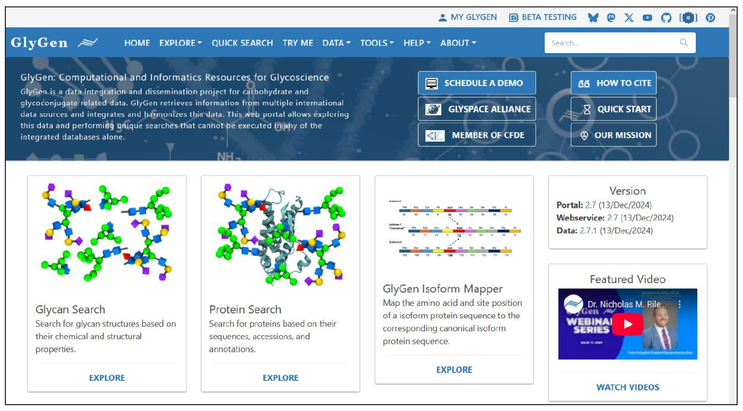
For each release data is downloaded from genomics, glycomics and proteomics databases. This data provided in different file formats and representations is parsed, extracted information is translated into standardized representations before integration into the GlyGen data model. Part of this data harmonization step is rigorous quality control. Downloaded and integrated datasets are versioned and archived in our data portal as CSV files.

Organisms in GlyGen



Data access

Web portal - ***https://www.glygen.org***

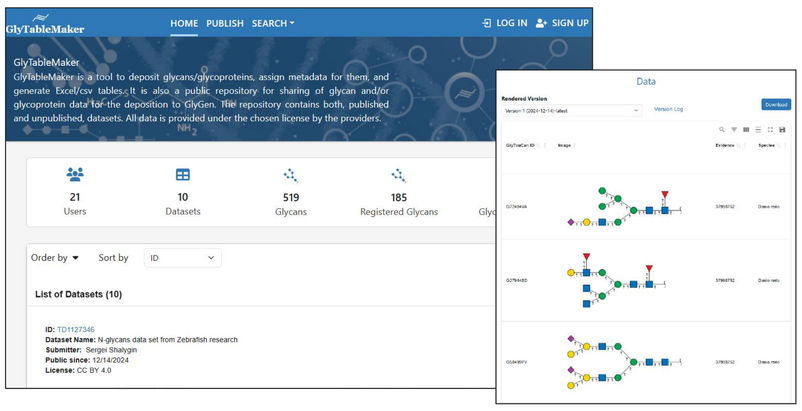


Datasets · API · SPARQL

https://data.glygen.org ***https://api.glygen.org*** ***https://sparql.glygen.org***

GlyTableMaker

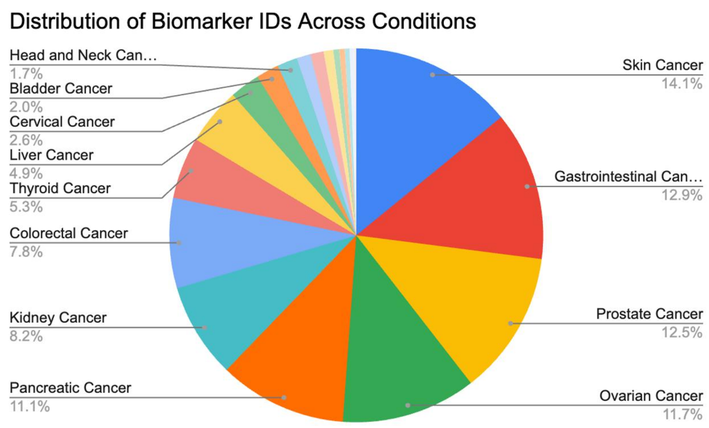
GlyTableMaker is a glycomics/glycoproteomics curation tool and data repository. Curators and data generators can use the online platform to add glycan/glycoproteins and their associated meta-data. After public release the data is freely avaiable in machine readable formats using standardized dictionairies and onotlogies for the data representation.



https://glygen.ccrc.uga.edu/tablemaker/

BiomarkerKB

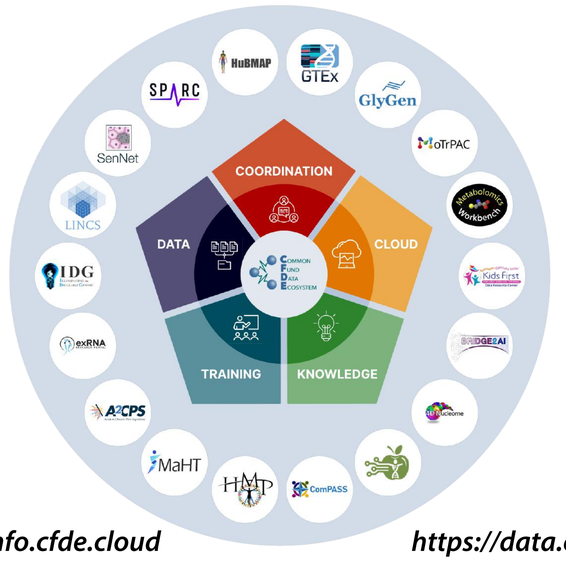
BiomarkerKB is a knowledgebase for molecular biomarkers, their conditions and associated metadata. Currently, the knowledgebase hosts over 132,000 unique biomarkers, including 120,000 gene biomarkers, 300 protein biomarkers, and 72 metabolite biomarkers.



https://biomarkerkb.org

Common Fund Data Ecosystem (CFDE)

CFDE aims to enable broad use of Common Fund datasets, and to promote data FAIRness and data sharing. GlyGen data is integrated into the CFDE workbench and CFDE metadata search portal.



https://info.cfde.cloud ***https://data.cfde.cloud***

Data license
Creative Commons CC BY 4.0

Software license
GNU General Public License v3

Software code on GitHub
https://github.com/glygener/

Papers

GlyGen: Computational and Informatics Resources for Glycoscience.
York et al. (2020) Glycobiology 30(2):72-73; PMID: 31616925.

GlyGen data model and processing workflow.
Kahsay et al. (2020) Bioinformatics 36(12):3941-43; PMID: 32324859

GlyGen is supported by the NIH Common Fund Glycoscience Program [U01GM125267], NIGMS [1R24GM146616-01], and NIH Office of Strategic Coordination [OT2 OD032092-01], [1U24OD038423-01].